Hermite–Biehler and the Pandrosion Normal Field

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April 2026

Abstract

We prove the Hermite–Biehler theorem using the imaginary part of the Pandrosion field. For $P(z) = \prod(z - \alpha_j)$ with all roots in the upper half-plane ($\mathrm{Im}(\alpha_j) > 0$), the "normal component" of the Pandrosion field on the real axis is $\mathrm{Im}(P'/P) = \sum y_j/|x - \alpha_j|^2 > 0$. This forces $\arg P(x)$ to be strictly increasing, which implies the interlacing of the real and imaginary parts A and B of P on the real line. The Wronskian $W(A, B) = -|P|^2 \cdot \mathrm{Im}(P'/P) < 0$ has constant sign, providing a direct link to Sturm's interlacing criterion.

1 The Hermite–Biehler theorem

Theorem 1.1 (Hermite–Biehler)**.** *A polynomial $P(z)$ has all roots in the open upper half-plane $\mathbb{H}^+ = \{z : \mathrm{Im}(z) > 0\}$ if and only if $P(x) = A(x) + iB(x)$ on the real axis, where A and B are real polynomials whose real zeros strictly interlace.*

2 The Pandrosion normal field

Definition 2.1. The *Pandrosion field* of P is $F_P(z) = P'(z)/P(z) = \sum_j 1/(z - \alpha_j)$. On the real axis $z = x$, it decomposes into tangential and normal components:

$$\mathrm{Re}\, F_P(x) = \sum_j \frac{x - x_j}{|x - \alpha_j|^2}, \tag{1}$$

$$\mathrm{Im}\, F_P(x) = \sum_j \frac{y_j}{|x - \alpha_j|^2}. \tag{2}$$

where $\alpha_j = x_j + iy_j$.

Lemma 2.2 (Normal field positivity)**.** *If all $y_j > 0$ (roots in $\mathbb{H}^+$), then:*

$$\boxed{\mathrm{Im}\frac{P'(x)}{P(x)} = \sum_{j=1}^{n} \frac{y_j}{|x - \alpha_j|^2} > 0 \qquad \forall x \in \mathbb{R}.} \tag{3}$$

3 Proof of Hermite–Biehler

Pandrosion proof of Theorem 1.1. ($\Rightarrow$) Suppose all $\alpha_j \in \mathbb{H}^+$.

Step 1: Normal field. By Lemma 2.2, $\mathrm{Im}(P'/P) > 0$ on $\mathbb{R}$.

Step 2: Argument rotation. Since $\frac{d}{dx} \arg P(x) = \mathrm{Im}\frac{P'(x)}{P(x)} > 0$, the argument $\arg P(x)$ is strictly increasing on $\mathbb{R}$. The total increase from $x = -\infty$ to $x = +\infty$ is $n\pi$.

Step 3: Interlacing. Writing $P(x) = A(x) + iB(x)$: the zeros of A occur where $\arg P = \pi/2 + k\pi$, and zeros of B where $\arg P = k\pi$. Since $\arg P$ is strictly increasing, these are interleaved: between consecutive zeros of A lies exactly one zero of B, and vice versa.

Step 4: Wronskian. $W(A,B) = A'B - AB'$. One computes:

$$W(A,B) = -\mathrm{Im}(\overline{P} \cdot P') = -|P|^2 \cdot \mathrm{Im}\frac{P'}{P} < 0. \tag{4}$$

The constant sign of $W(A,B)$ is equivalent to interlacing (Sturm's criterion).

($\Leftarrow$) If A and B interlace, then $W(A,B) \neq 0$ on $\mathbb{R}$, so P has no real roots. The winding number of P along the real axis equals $n\pi$, forcing all n roots into one half-plane. Since $\mathrm{Im}(P'/P)$ has constant sign, it must be positive, so all roots have $y_j > 0$. $\square$

4 Connection to the Wronskian calculus

Proposition 4.1. *From Paper 64: $W(P, Q_j) = Q_j^2$. For P with roots in $\mathbb{H}^+$:*

$$W(A,B) = -|P|^2 \cdot \mathrm{Im}\, F_P = -|P|^2 \sum_j \frac{y_j}{|x-\alpha_j|^2}. \tag{5}$$

Each summand $y_j/|x-\alpha_j|^2$ is the fiber contribution *of root α_j to the Wronskian. The interlacing is driven by the "downward pressure" of each root on the Wronskian.*

5 Numerical verification

Property	**Tests**	**Violations**
A has all real roots	10 000	0
$\mathrm{Im}(P'/P) > 0$ on $\mathbb{R}$	10 000	0
$\mathrm{Im}(P'/P) = \sum y_j/\lvert x-\alpha_j\rvert^2$	5 000	err $< 10^{-13}$
$W(A,B)$ constant sign	10 000	0
$W(A,B) = -\mathrm{Im}(\bar{P}P')$	1 000	1000/1000
$\arg P(x)$ monotone increasing	5	all confirmed
Interlacing of A, B roots	5	all confirmed

6 The Pandrosion perspective

Component	**Formula**	**Meaning**
Tangential	$\mathrm{Re}(P'/P)$	speed of zero dynamics
Normal	$\mathrm{Im}(P'/P)$	half-plane confinement
Wronskian	$W(A,B) = -\lvert P\rvert^2\,\mathrm{Im}(P'/P)$	interlacing force
Turán (Paper 56)	$P'^2 - PP'' = \sum Q^2$	real-rootedness

The Pandrosion field is the unifying object: its real part governs the dynamics (Rolle, Sturm), its imaginary part governs the geometry (Hermite–Biehler), and its modular square governs the algebra (Turán SOS).

References

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